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324513 (24)

BE (5th Semester)

Examination, April - May, 2013

Branch : Electrical

ANALOG ELECTRONICS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Part (a) of each question is compulsory.

(ii) Attempt any two parts from (b), (c) & (d).

(iii) Assume suitable data if necessary.

Unit - I

Q. 1. (a) Draw the Hybrid-π model as well as h-parameter model for common emitter configuration at low frequency. 2

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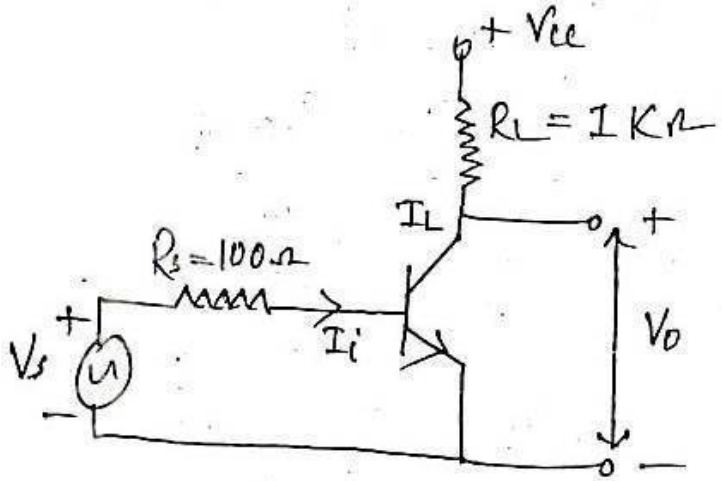
(b) The hybrid-π parameters of the transistor used in the circuit of below figure are given as 7

g_m = 50 mA/V, $r_{b'e}$ = 1 kΩ, $r_{b'c}$ = 4 MΩ,

r_{ce} = 80 kΩ, C_c = 3 pF, C_e = 100 pF,

$r_{bb'}$ = 100 Ω.



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Calculate :

(i) Upper 3 dB frequency of current gain

$A_i = \frac{I_L}{I_i}$

(ii) Magnitude of voltage gain at

$A_{vs} = \frac{V_o}{V_s}$ at frequency of part (i).

(c) A bipolar junction transistor has the following parameters measured at I_C = 1 mA; 7

h_{ie} = 3kΩ, h_{fe} = 100, f_T = 4 MHz, C_c = 2 pF

C_e = 18 pF.

Calculate : (i) $r_{b'e}$

(2) $r_{bb'}$

(3) g_m

(4) f_H for R_L = 1 kΩ

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(d) Derive an expression for transconductance g_m ? Also explain its relationship with the temperature ? 7

Unit - II

Q. 2. (a) What are the advantages of pushpull amplifier over a single transistor amplifier ? 2

(b) List out the advantages and disadvantages of class A and class B Amplifier ? 7

(c) Explain class A power amplifier ? Derive the expression for efficiency and find the maximum efficiency ? (For transformer coupled class A Amp'). 7

(d) A class B pushpull amplifier drives a load of 16 Ω, connected to the secondary of the

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ideal transformer. The supply voltage is 25 Volt. If the number of turns on the primary is 200 and the number of turns on secondary is 50. 7

Calculate :

(1) Maximum power o/p

(2) DC power input

(3) Efficiency

(4) Maximum power dissipation per transistor

Unit - III

Q. 3. (a) What is the effect of coupling, bypass and spurious shunting capacitors on freq. response ? 2

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(b) Explain the effect of cascading Bandwidth, Rise time and Gain.

(c) An amplifier consists of 3 identical stages cascade, the bandwidth of overall amplific extends from 20 Hz to 20 KHz. Calculate the bandwidth of individual stage. 7

(d) Define and explain : 7

(1) Frequency distortion

(2) Phase shift distortion

(3) Thermal noise

Unit - IV

Q. 4. (a) Draw the symbolic representation of OP-Amp with indicating inverting and non-inverting inputs ? 2

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(b) Define and explain : 7

(1) Power Supply Rejection Ratio

(2) Slew Rate

(3) Error Voltage

(4) CMRR

(c) Explain any two basic application of OP-Amp ? 7

(d) Define and explain : 7

(1) Input Offset Voltage

(2) Input Bias Current

(3) Virtual Ground Concept

(4) Input Offset Current

Unit - V

Q. 5. (a) What are the advantages offered by integrated circuit technology as compared

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with discrete components interconnect o printed circuit board ? 7

(b) Write a short note, on photolithography and explain the process involved in it ? 7

(c) Explain the different steps involved in fabrication of integrated circuit with neat and clean figures ? 7

(d) Briefly explain the process of Ion Implantation ? 7

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